

PRANAV PANKHAWALA

AI Automation & Cybersecurity Engineer

CONTACT

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SKILLS

Cybersecurity & Hacking

- Cyber Security, Red Teaming, Vulnerability Scanning, Social Engineering, OWASP, Ethical Hacking, Penetration Testing, Network Security

AI & Machine Learning

- Artificial Intelligence, Machine Learning, Computer Vision

CV & Frameworks

- OpenCV, YOLO, Pose Estimation, Media Pipe, PyTorch

Networking & IoT

- Internet of Things, MQTT

OS & Platforms

- Linux, Kali Linux, Universal Windows Platform

Web & API Dev

- Fast API, Flask

Programming

- C/C++, Python

EDUCATION

MTech – CSE

Network Mgmt. & Cyber Security
MIT WPU | 2022 | 9.14/10

BE – E&TC

SPPU | 2020 | 7.36/10

Diploma – Cyber Law

Asian School of Cyber Laws
2023 | 62%

XII – Science

Sinhgad Spring Dale Public School |
2016 | 71.6%

X – SSC

Sinhgad Spring Dale Public School |
2014 | 70.8%

OBJECTIVE

Seeking a challenging role in Artificial Intelligence, Computer Vision, and Cybersecurity to develop secure, scalable, and real-time intelligent systems for industrial automation, smart manufacturing and autonomous driving.

ABOUT

Innovative and results-driven AI and Cybersecurity professional with extensive experience at the intersection of Artificial Intelligence, Computer Vision, and Industrial Cybersecurity. Specialized in designing, developing, and deploying intelligent, secure, and scalable machine learning systems. Proven expertise in CNNs and YOLO-based architectures for defect detection, object classification, and automated inspection across manufacturing workflows. Skilled in edge computing platforms (NVIDIA Jetson Nano, AGX Xavier, Raspberry Pi) and model optimization. Strong cybersecurity acumen with formal training in cyber law and secure system design. Proficient in PyTorch, OpenCV, Fast API, and Flask. Continuously updating expertise through certifications in AI, cybersecurity, and generative AI.

EXPERIENCE

AI Automation Engineer

C4i4 Lab | Jul 2025 – Present

- Designed and deployed AI-powered defect detection and object classification systems for manufacturing quality assurance.
- Built and optimized deep learning models (CNNs, YOLO) for real-time object detection, surveillance, and industrial inspection.
- Enhanced detection accuracy via colour-based rejection algorithms, Canny edge detection, contour analysis, and preprocessing optimization.
- Deployed inference pipelines on NVIDIA Jetson Nano, Jetson AGX Xavier, and Raspberry Pi for low-latency processing.
- Optimized models via dataset augmentation, hyperparameter tuning, model quantization, and hardware-aware strategies.
- Developed end-to-end CV workflows and evaluated performance using precision, recall, F1-score, and inference latency.

Engineer Intern → Software Developer

C4i4 Lab | Sep 2023 – Jul 2025

- Contributed to QUIKSTEP software for a 0.5HP water pump demonstration line, improving operational reliability and efficiency.
- Designed scalable backend services and RESTful APIs using Django and Python; built responsive frontends with HTML/CSS/JS.
- Integrated AI-assisted development tools, reducing development time by approximately 30%.
- Progressed from Engineer Intern to Software Developer through consistent technical contributions and collaboration.

Cybersecurity Engineer Intern

Photoshooto | Jan 2022 – Mar 2022

- Conducted VAPT using TryHackMe and industry-standard methodologies to strengthen offensive security capabilities.
- Identified, analysed, and documented vulnerabilities with detailed risk assessments and remediation recommendations.
- Performed reconnaissance, enumeration, exploitation, and post-exploitation tasks to evaluate system security posture.
- Promoted from intern to team lead; coordinated small team under tight deadlines ensuring successful delivery.

LANGUAGE PROFICIENCY

English – Fluent
Hindi – Fluent
Marathi – Native
Gujarati – Conversational
Japanese – Elementary

INTERESTS / HOBBIES

Motorcycle Riding
Gym
Music
Bug Bounty / Ethical Hacking

PERSONAL DETAILS

Marital Status: Single

Engineer Intern

Twintech Control System | Dec 2018 – Jan 2019

- Developed a C# .NET UWP application for Windows 10 IoT Core on Raspberry Pi to monitor industrial thermal switch systems.
- Built real-time temperature monitoring, alert management, and threshold configuration functionalities.
- Integrated sensor data acquisition and communication protocols for accurate environmental monitoring.

PROJECTS

AI-based Surveillance System *(Self / Ongoing)*

Developing an AI-based surveillance system for traffic violation detection, industrial surveillance, and human security.

ML-based Network Security Model *(MTech)*

Developed an AI/ML model for detecting malicious network packets based on behavioural markers.

Night Vision & Perimeter Security *(BE)*

Built a night vision system using OpenCV and an unarmed vehicle for perimeter security monitoring.

PUBLICATIONS

Bankar M., Budukh A., Pankhawala P. 'Leveraging Potential of Industrial 4.0: A Comprehensive Analysis of Smart Manufacturing Technologies and Applications.' IPE Journal of Management, ISSN 2249-9040, Vol. 14, No. 2, Jan–Jun 2024.

Pankhawala P. & Dr. Sirsikar S. 'Machine Learning Based Network Security in Internet of Things.' Neuro Quantology, DOI: 10.14704/nq.2022.20.9. NQ440121.

CERTIFICATIONS

- **AI for Cyber Security** — LinkedIn Learning (May 2026)
- **OpenCV Bootcamp** — OpenCV University (Feb 2026)
- **Vision Language Models (VLM) Bootcamp** — OpenCV University (Feb 2026)
- **Python using AI** — AI for Techies (Aug 2025)
- **Fundamentals of Information Security** — Infosys (Mar 2023)
- **Learn Python and Ethical Hacking from Scratch** — Udemy (Feb 2023)
- **C Tutorial Course** — Solo Learn (Jun 2019)
- **Introduction to Programming Using Python** — Udemy (Jan 2019)
- **Shell Scripting** — Udemy (Dec 2018)
- **SQL Fundamentals** — Solo Learn (Oct 2018)
- **Introduction to Programming Using Java** — Solo Learn (Aug 2018)
- **C Programming** — SEED Infotech Ltd (Aug 2014)